

3 ATTENTION-BASED NEURO-SYMBOLIC RULE EXTRACTOR ARCHITECTURE

ANDRE learns logical rules from structured examples by introducing a rule space that enables smooth and interpretable rule induction. It also employs attention-based conjunction and disjunction mechanisms that approximate symbolic reasoning operations. Building on these components, a learning strategy is developed to discover subrules sequentially, followed by a detailed procedure for symbolic rule extraction after training.

3.1 RULE SPACE CONSTRUCTION

Since the rule and subrule structures vary dynamically in ILP, it is highly challenging to develop a universal, systematic neural network-based approach for identifying subrules for a given head predicate. This challenge is more akin to structure induction or representation learning in natural language processing, where models must infer latent semantic and syntactic relationships from variably structured prompts. To address this issue, a zero-padding technique is typically employed to enforce a fixed-size input compatible with neural networks. Adopting a similar technique in ILP mitigates the difficulty caused by the dynamic transformation of logical subrules, enabling a fixed structure suitable for neural modeling. Just as adding zero to neural inputs has a neutral effect, adding one to a logical conjunction also has no effect, due to the logical *identity property* ($1 \wedge b = b$). Based on this property, we represent the exclusion of B_{ij} from subrule h_i in Eq. 3 using the symbolic subpredicate 1. Thus, each B_{ij} can take only one of three possible symbolic subpredicates: b_j , $\neg b_j$, or 1. Here, b_j denotes the positive form of the predicate, $\neg b_j$ its negation, and 1 the absence of the predicate.

Following this strategy, we reformulate the ILP problem by converting Eqs. 2 and 3 into a matrix representation. Let the matrix $B^{n \times m}$ represents m body predicates across n subrules. As discussed, each array B_{ij} is:

$$B_{ij} = \{b \mid b \in \{b_j, \neg b_j, 1\}\}. \quad (4)$$

Let $S_j = \{b_j, \neg b_j, 1\}$; then $B_{ij} \in B$ refers to the target symbolic predicate to be determined. Let b_j , $1 - b_j$, and 1 be the valuations of b_j , $\neg b_j$, and 1, respectively. There are $3^m \times n$ possible combinations of subpredicates for B , making the rule space incredibly large. The challenge is to navigate this large discrete space using gradient-based optimization to recover the desired B given a set of examples E containing all b_j truth values and corresponding binary outputs.

To overcome this, we convert the discrete space into a continuous one and apply gradient-based methods to learn optimal subpredicates. Specifically, we define an adaptive probability distribution over the subpredicates in S_j to determine the most likely option. A trainable weight vector $\mathcal{W}_{ij} \in \mathbb{R}^3$ is assigned to each S_j , which is then passed through a softmax function to produce a probability distribution $P_{ij} \in [0, 1]^3$, where $P_{ij} = \{p_{ij}^{(1)}, p_{ij}^{(2)}, p_{ij}^{(3)}\}$ and $\sum_{k=1}^3 p_{ij}^{(k)} = 1$.

Let $S_j = \{p_{ij}^{(1)} b_j, p_{ij}^{(2)} (1 - b_j), p_{ij}^{(3)}\}$ be the set of weighted valuations for the subpredicates in S_j (illustrated in Figure 1). The goal is to amplify the most significant subpredicate while suppressing the others. While a hard max operator would select the largest value in S_j , it introduces non-differentiability. Instead, we use a differentiable approximation based on the probabilistic sum (prob-sum), which emphasizes the dominant input among probabilistic values (see Appendix A for further details):

$$f(x, y, z) = x + y + z - xy - xz - yz + xyz, \quad (5)$$

where $f : [0, 1]^3 \rightarrow [0, 1]$ is the prob-sum activation function applied to $x, y, z \in [0, 1]$. Using this function, the valuation of B_{ij} is computed as:

$$B_{ij} = f(p_{ij}^{(1)} b_j, p_{ij}^{(2)} (1 - b_j), p_{ij}^{(3)}), \quad (6)$$

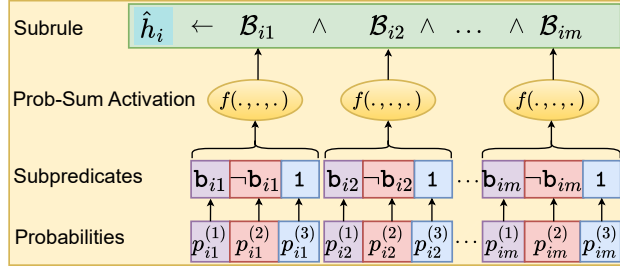


Figure 1: **Graphical representation** of a logical subrule structure within the context of the rule space.

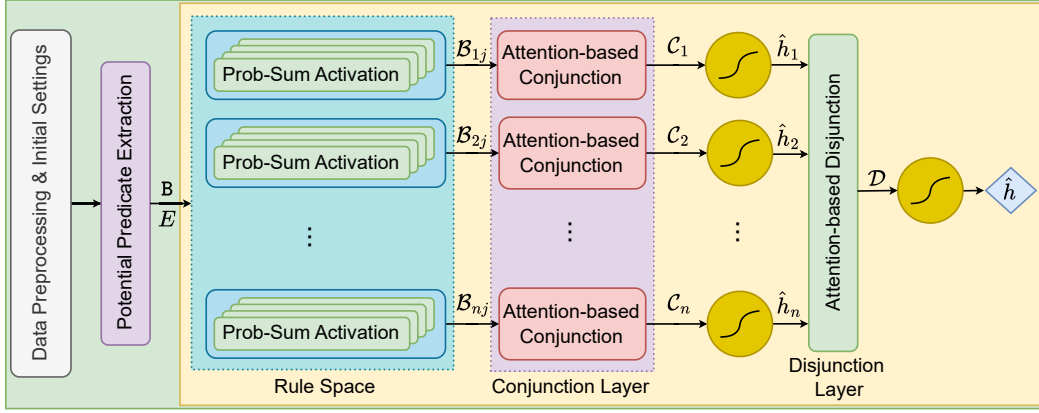


Figure 2: **Overview of ANDRE’s architecture.** ANDRE includes an innovative rule space with an *Attention-based Conjunction-Disjunction Network*.

where $\mathcal{B}_{ij}$ denotes the soft valuation of B_{ij} . This formulation quantifies the combined contribution of the three symbolic subpredicates and emphasizes the most dominant one.

Figure 1 visually demonstrates how the probabilities are assigned and then aggregated via Eq. 5 to compute $\mathcal{B}_{ij}$. This step is essential for constructing the continuous rule space. The overall objective is to efficiently explore this parameterized space and discover a set of subrules that accurately explain the positive and negative examples provided in the training set.

3.2 LOGICAL NETWORK OF ANDRE

Having constructed the continuous rule space with trainable probabilities, we now describe ANDRE’s logical network. Following the computation of all body predicate valuations, ILP systems typically apply a logical conjunction-disjunction mechanism to infer the value of the head predicate. Recently, Hu et al. (2025) demonstrated that a single softmax attention layer can approximate k -Boolean functions (e.g., AND/OR). Inspired by this work, we adopt and customize a new strategy to approximate hard minimum and maximum operators using soft attention.

Conjunction Layer. As shown in Figure 2, after constructing the rule space and computing $\mathcal{B}_{ij}$ for subrule h_i , the valuation of the subrule head is determined by aggregating the values of all $\mathcal{B}_{ij}$ through a conjunction operation. The attention-based conjunction is calculated as follows:

$$\mathcal{C}_i = \sum_{j=1}^m \frac{e^{s_{\min}(\mathcal{B}_{ij})}}{\sum_{k=1}^m e^{s_{\min}(\mathcal{B}_{ik})}} \mathcal{B}_{ij}, \quad (7)$$

where $\mathcal{C}_i \in [0, 1]$ represents the soft conjunction of the body predicates in subrule h_i . This mechanism uses adaptive attention scores $s_{\min} : [0, 1] \rightarrow (-\infty, 0]$, which assign higher attention to smaller values:

$$s_{\min}(\mathcal{B}_{ij}) = -\beta \mathcal{B}_{ij}, \quad (8)$$

where $\beta \in (0, \infty)$ is a hyperparameter controlling the sharpness of the attention distribution. Higher values of β improve the approximation to a hard minimum but may introduce instability. Eq. 7 yields a weighted average of the $\mathcal{B}_{ij}$ terms, where the weights are given by a softmax over the negative attention scores. Figure 3 illustrates how each $\mathcal{C}_i$ is computed using the attention-based conjunction.

Since the output of each subrule is expected to be binary, a sigmoid activation function is applied to convert $\mathcal{C}_i$ into a value close to 0 or 1: $\hat{h}_i = \text{sigmoid}(\lambda(\mathcal{C}_i - \gamma))$, where $\lambda \in (0, \infty)$ controls the steepness of the sigmoid function and $\gamma \in (0, 1)$ determines its midpoint. This process is repeated for each subrule h_i , yielding predicted outputs $\hat{h}_i \in [0, 1]$, which are then used in the disjunction step.

Disjunction Layer. Once all subrule outputs $\hat{h}_i$ are computed, they must be combined to infer the overall head predicate value, as shown in Figure 2. To do this, we apply a similar attention-based

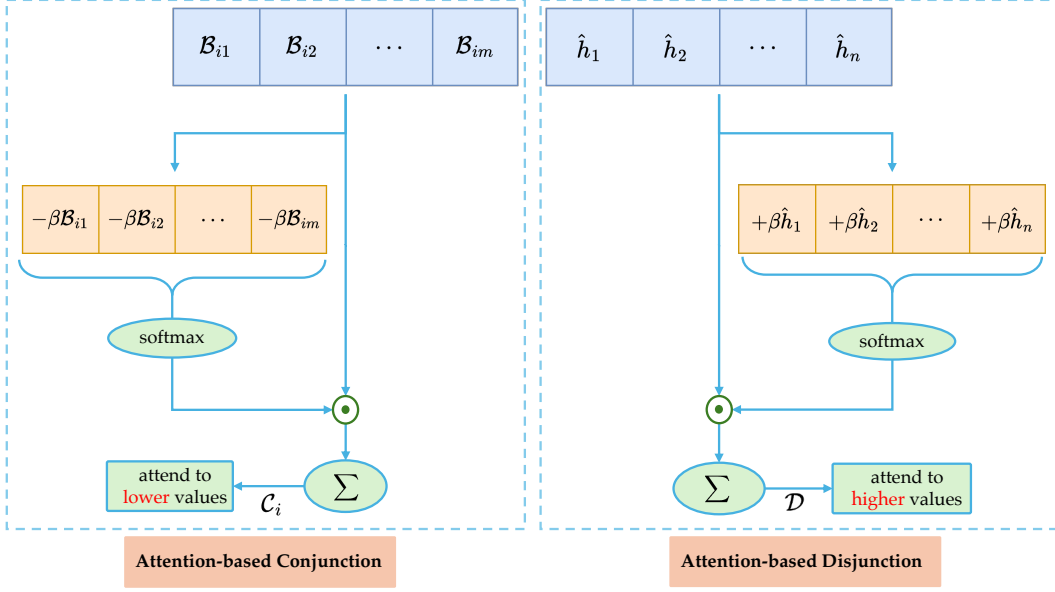


Figure 3: Graphical representations of attention-based conjunction and disjunction operators. Softmax with $-\beta$ represents softmin attention (conjunction), while softmax with $+\beta$ stands for softmax attention (disjunction). Symbol $\odot$ indicates a dot-product operator.

strategy that approximates the maximum among the subrule outputs. The disjunction is computed as:

$$\mathcal{D} = \sum_{i=1}^n \frac{e^{s_{\max}(\hat{h}_i)}}{\sum_{k=1}^n e^{s_{\max}(\hat{h}_k)}} \hat{h}_i, \quad (9)$$

where $\mathcal{D} \in [0, 1]$ represents the soft disjunction of all subrule outputs. The attention scores $s_{\max} : [0, 1] \rightarrow [0, \infty)$ are defined in opposition to those in Eq. 8: $s_{\max}(\hat{h}_i) = -s_{\min}(\hat{h}_i)$. This formulation emphasizes higher-valued subrules, allowing the model to approximate a logical OR operation. Figure 3 indicates how $\mathcal{D}$ is computed using an attention-based disjunction. As with conjunction, a sigmoid activation is applied to map the disjunction output to a binary value: $p(\hat{h} | e, \mathcal{W}) = \text{sigmoid}(\lambda(\mathcal{D} - \gamma))$, where $p(\hat{h} | e, \mathcal{W}) \in [0, 1]$ is the final predicted probability of the head predicate for input example e with ANDRE’s weight configuration $\mathcal{W} \in \mathbb{R}^{n \times m \times 3}$, which includes all subpredicate weights $\mathcal{W}_{ij}$.

Figure 2 illustrates the complete ANDRE architecture, including the construction of the rule space and the attention-based conjunction and disjunction layers. This design enables symbolic rule extraction through a fully differentiable neural structure that learns logical dependencies directly from binary-labeled examples. Moreover, Appendix B mathematically proves that the proposed attention-based operators approximate hard minimum and maximum operators more accurately compared to product-based t -norms and s -norms on probabilistic data.

3.3 OPTIMIZATION

Given the valuations of all potential body predicates and corresponding Boolean outputs, ANDRE is designed not only to identify the most relevant subpredicates but also to ensure that the resulting rules adhere to valid logical formats. To achieve this, the learning process is guided by two complementary objectives: *semantic learning* and *syntactic learning*.

Semantic Learning. This learning paradigm aims to learn a set of optimal weights $\mathcal{W}$ that enable the network to accurately predict the Boolean outputs while highlighting the most confident subpredicates. It involves optimizing three general loss functions:

Binary Cross-Entropy Loss: This loss function evaluates the confidence of each prediction. By optimizing this loss, ANDRE learns to predict Boolean outcomes for each sample, ensuring consistency